Assembly — VX-220 Servo Drive Controller

Line 2 · Stations ST-40 and ST-70 · Part number NVE-VX220-A3

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Applies to	Stations ST-40 and ST-70, benches WS-401 to WS-403 · Product NVE-VX220-A3 · Shifts A and B
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1. Purpose

This procedure defines how the VX-220 Servo Drive Controller is assembled at station ST-40: how the operator takes up a released work order, the sequence in which the components are brought together, the fastening torques, how completed quantity is recorded against the order, and what is printed on the unit label.

2. Scope

Applies to part number NVE-VX220-A3 at stations ST-40 and ST-70, benches WS-401 to WS-403, on shifts A and B. Assembly and fastening are performed at ST-40; labelling and packing at ST-70.

The enclosure and heat-spreader block are machined at ST-20, deburred and cleaned at ST-30, and reach this station from ST-30. The populated PCBA is transferred in from Plant 1. Functional test at ST-50 is performed to the FCT-401 operating instructions and is outside this procedure. Visual inspection at ST-60 is covered by NVE-SOP-102. Machine start-up verification at ST-20 is covered by NVE-FRM-012.

3. Prerequisites

- The operator is trained and signed off on this procedure and on ESD handling.
- The bench is at an ESD-protected workstation. The wrist strap has been tested at the start of shift and read within 0.8 to 35 MΩ.
- An electric torque driver TDR-3000 is present at the bench and within its calibration date, with both programmes for this product available: **P1 at 2.5 N·m** and **P2 at 0.6 N·m**.
- A kit for the work order has been staged at ST-10 against NVE-BOM-2201. The kit contains every item in Table 1: the enclosure and block arrive at ST-10 from ST-30 and are kitted with the purchased items, so the operator draws all parts from the kit and none from the line.

4. Components

Table 1 — Components consumed per unit. Full detail in NVE-BOM-2201 Rev 04.

Part number	Description	Qty
NVE-ENC-2201	Enclosure, milled aluminium 6082-T6, externally finned	1
NVE-HSK-2204	Heat-spreader block, milled aluminium 6082-T6	1
NVE-PCB-2210	PCBA, servo control board, populated	1
NVE-TIM-0045	Thermal interface pad, 40 × 40 × 1.0 mm	1

Part number	Description	Qty
NVE-FST-M412	Screw, socket head M4 × 12, stainless	4
NVE-FST-M308	Screw, socket head M3 × 8, stainless	3
NVE-CVR-2207	Connector cover, moulded PA66	1
NVE-LBL-2220	Product label, 50 × 25 mm	1

5. Taking up a work order

Work orders are released to Line 2 by Production Planning and appear at the station in released status. The operator selects one released work order and starts the assembly operation against it. Only one work order is worked at a time per bench.

On starting, the following are recorded against the order: the operator identity, the start date and time, and the station. A batch identifier is assigned to the order at first start and stays with it for the life of the order.

Work order: _____ Product: _____ Quantity required: _____ pcs
Batch ID: _____ Operator: _____ Bench: _____

6. Assembly sequence

The assembly is performed in **three steps**, in the order given. Each step, including its fastening, is completed in full before the next is started. A reference image and a short instruction video are available at the bench for each step.

> **How the unit gets rid of its heat.** The board's power devices clamp against the upper face of the heat-spreader block. The block sits on the thermal pad, and the pad passes the heat into the enclosure, which carries the fins on its outside. The path is **devices → block → pad → enclosure → ambient**. Every step below protects one link in that path: a folded pad, a block that does not sit flat, or a board not pulled down onto the block all break it.

Step 1 — Fit the thermal pad and heat-spreader block

1. Place the enclosure NVE-ENC-2201 in the bench fixture, connector cut-out towards you.
2. Peel the liner from one face of the thermal pad NVE-TIM-0045.
3. Lay the pad flat in the centre of the enclosure base, between the four screw bosses. The pad must sit flat, fully inside the bosses, and must not ride up a wall.
4. Peel the remaining liner.
5. Place the block NVE-HSK-2204 on the pad, its machined upper face uppermost, and align its four holes with bosses 1 to 4.
6. Press down lightly by hand until the block sits flat on the pad.
7. Fit the four M4 screws NVE-FST-M412 through the block into the bosses. Start each one by hand until it takes the thread. Do not drive a screw that will not start by hand.
8. Select driver programme **P1**. Tighten the four screws to **2.5 N·m** in the order **1-3-2-4**. Boss numbering is on the drawing: 1 front-left, 2 front-right, 3 rear-right, 4 rear-left, viewed with the connector cut-out towards you.
9. Check the driver signalled a good fastening on all four screws — **green lamp and a single beep**. A red lamp and a double beep is a fault.

Reject: a pad that is creased, folded, torn, or pressed down off-centre. Peel it off, discard it, wipe the base clean of adhesive residue with an isopropanol wipe, and fit a new pad — do not reposition a pad that has already been pressed down.

Step 2 — Fit the PCBA

1. Take the PCBA NVE-PCB-2210 from the ESD-safe tray by its edges.
2. Check the revision marking on the board reads **A3**.
3. Lower the board in at an angle, connector end first, so the connector body enters the cut-out.

4. Bring the board down flat, so that its power devices rest on the upper face of the block and the board meets all three standoffs.
5. Fit the three M3 screws NVE-FST-M308 through the board into standoffs 1 to 3, front to rear, starting each by hand.
6. Select driver programme **P2**. Tighten the three screws to **0.6 N·m** in the order **1-2-3**. These screws also set the contact pressure between the devices and the block, so all three must reach torque.
7. Check the connector body is flush with the outside of the enclosure, and that no gap is visible at any standoff.

Reject: a board marked **A2**. The A2 board carries its power devices on the opposite edge and cannot reach the block on this enclosure. Return it to ST-10 — do not attempt to fit it.

Step 3 — Fit the connector cover

The board is now fastened, so the cover can be pressed on without moving it.

1. Hold the connector cover NVE-CVR-2207 over the connector, cable exit downwards.
2. Press the cover on until both clips click.
3. Pull the cover gently by hand. It must not lift.

Reject: a cover that lifts when pulled, or that will not click on both clips. Take it off, check both clips for damage, and fit a new cover if either is damaged.

☐ All seven fasteners confirmed good Driver faults this unit: _____ Recorded in the fastening fault log: ☐ Yes ☐ None to record

> **Fastening is not a separate later step.** The four M4 screws sit underneath the board, so they are tightened in Step 1, before the board goes in. Between placing the block and tightening those screws, the block is held only by its own weight — do not lift or tilt the enclosure until Step 1 is complete.

7. Fastening specification

The fastening actions are in Steps 1 and 2. This section is the specification they work to, and the rules that apply to both groups.

Table 2 — Fastening specification

Fastener	Joint	Prog.	Torque	Tolerance	Sequence
NVE-FST-M412 (× 4)	Block to enclosure, into tapped boss	P1	2.5 N·m	± 0.2 N·m	1-3-2-4, diagonal
NVE-FST-M308 (× 3)	PCBA to enclosure, into tapped standoff	P2	0.6 N·m	± 0.05 N·m	1-2-3, front to rear

- **Numbering.** Bosses 1 to 4 and standoffs 1 to 3 are marked on drawing NVE-DWG-2201 sheet 3 and etched beside each hole in the enclosure. Viewed with the connector cut-out towards the operator, boss 1 is front-left, 2 front-right, 3 rear-right, 4 rear-left.
- **One programme per group.** P1 for the M4 group, P2 for the M3 group. The driver does not change target within a group, and a group is never driven on the other group's programme.
- **Good fastening** is a green lamp and a single beep. A fault is a red lamp and a double beep.
- **Tighten each screw once only.** Do not re-tighten a screw that has already reached torque, and do not finish a screw with a hand tool.

Reject: a driver fault on any screw — torque not reached, or torque reached early. Stop, do not proceed, and raise the unit to the Line Lead. Record the unit, the screw position and the fault in the fastening fault log. A stripped boss is a scrap condition for the enclosure.

Programme P1 used: ☐ Programme P2 used: ☐ Screw positions with a fault: _____

8. Recording completed quantity

> **"Batch" means one thing in this procedure:** the whole work order, identified by the Batch ID assigned at first start. A tray holds ten units and is a handling quantity, not a batch.

Completed units are counted against the work order as they are finished. The operator increments the completed quantity by one for each good unit, or by ten when closing out a full tray of ten. The count may be corrected downwards if a unit is subsequently rejected at ST-50 (functional test) or ST-60 (visual inspection) and cannot be reworked.

The station shows the quantity completed against the quantity required. The operation is closed only when the two agree, or when the Line Lead authorises a short close with a reason recorded.

Quantity completed: _____ pcs Quantity required: _____ pcs Short close authorised by: _____

9. Completion and labelling

Labelling is performed at ST-70. On closing the operation, confirm the completion record. The following are shown and must all be checked before any label is printed:

- Work order number
- Product
- Operator
- Batch ID
- Quantity completed — because it is printed on every label and has a reject criterion below

One label NVE-LBL-2220 is printed and applied per unit, on printer LP-601, to the outer face of the enclosure below the connector cut-out, square to the edge within 2 mm measured against the lower edge of the enclosure. The label carries exactly the following fields:

Table 3 — Label content, NVE-LBL-2220

Field	Content
Work order	Work order number the unit was built against
Batch ID	Batch identifier assigned to the work order at first start
Batch quantity	Quantity completed on the work order at the time of printing
Product	Part number and description
Operator	Operator who closed the operation
Date printed	Date the label was printed

Reject: a label that is unreadable, misaligned by more than 2 mm, lifted at any corner, or carrying a batch quantity that does not match the quantity completed at the time of printing. Destroy it, record the destruction in the label reprint log, and print a replacement.

Labels printed: _____ Labels destroyed and reprinted: _____ Entered in the reprint log: ☐ Yes ☐ None to record

> Where a unit is rejected downstream after its label was printed, the quantity completed drops but the labels already applied are not reprinted. The Batch ID, not the printed quantity, is what traces a unit to its order.

10. Records

Record	Retained by	Retention
Work order completion record	Production	7 years

Record	Retained by	Retention
Fastening fault record	Manufacturing Engineering	3 years
Label reprint log	Production	1 year

11. Related documents

- NVE-BOM-2201 Rev 04 — Bill of Materials, NVE-VX220-A3
- NVE-SOP-102 — Visual Inspection, VX-220
- NVE-SOP-104 — Operator Roles and Responsibilities, Line 2
- NVE-LST-001 — Equipment List, Line 2
- NVE-FRM-011 — End-of-Shift Report Template, Line 2
- TDR-3000 Torque Driver Operating Instructions, document TDR-OI-14

Revision history

Rev	Date	Changed by	Reason for change
01	2023-02-06	M. Halász	Initial issue for VX-220 revision A1.
02	2024-04-22	M. Halász	Revision A2 enclosure introduced. Heatsink fastener changed from M5 to M4.
03	2025-09-01	M. Halász	Field returns for thermal fault investigated. Root cause was inconsistent thermal pad compression from a worn driver clutch. Driver calibration interval shortened to 6 months and the torque tolerance stated explicitly as ± 0.2 N·m.
04	2026-05-11	M. Halász	Revision A3 enclosure. Assembly split into three defined steps. Quantity recording and label content specified in sections 8 and 9.
05	2026-08-17	M. Halász	Thermal path corrected: NVE-HSK-2204 is a heat-spreader block bearing on the pad, with the fins on the outside of the enclosure, and the M3 screws set the device-to-block contact pressure. Fastening moved into Steps 1 and 2, because the M4 screws sit under the board. Boss and standoff numbering, driver programmes P1 and P2, and the good-fastening signal specified. M3 torque reduced from 1.2 to 0.6 N·m to protect the laminate. Label corrected to one per unit.

Approvals

Function	Name	Signature	Date
Author	M. Halász Manufacturing Engineer		2026-07-27
Reviewer	P. Vargha Production Supervisor		2026-08-04
Approver	K. Antal Quality Manager		2026-08-11